

4-Bit Signed ALU - Truth Tables and Test Cases

This document lists binary test cases used to verify the full ALU. All inputs and outputs are written in binary. For ADD/SUB, overflow is the signed two's complement overflow flag, not Cout.

ADD Tests, SUB = 0

SUB	A	B	Expected Sum	Cout	Overflow
0	0000	0000	0000	0	0
0	0000	0001	0001	0	0
0	0001	0001	0010	0	0
0	0011	0010	0101	0	0
0	0111	0001	1000	0	1
0	0100	0100	1000	0	1
0	1111	1111	1110	1	0
0	1000	1111	0111	1	1
0	1010	1100	0110	1	1
0	0101	1011	0000	1	0

SUBTRACT Tests, SUB = 1

SUB	A	B	Expected Sum	Cout	Overflow
1	0000	0000	0000	1	0
1	0001	0001	0000	1	0
1	0011	0001	0010	1	0
1	0000	0001	1111	0	0
1	0101	0010	0011	1	0
1	0010	0101	1101	0	0
1	0111	1111	1000	0	1
1	0100	1100	1000	0	1
1	1000	0001	0111	1	1
1	1010	0011	0111	1	1

Overflow Flag

The overflow flag tells whether the 4-bit signed result is outside the valid range -8 to +7. Cout is not the signed overflow flag. Cout is only the carry out of the adder.

$B_eff3 = B3 \text{ XOR } SUB$
 $Overflow = NOT(A3 \text{ XOR } B_eff3) \text{ AND } (A3 \text{ XOR } S3)$

For subtraction only:
 $Overflow = (A3 \text{ XOR } B3) \text{ AND } (A3 \text{ XOR } S3)$

Signed Comparators Tests

A	B	GT	EQ	LT
0000	0000	0	1	0
0001	0000	1	0	0
0000	0001	0	0	1
0111	1111	1	0	0
1111	0111	0	0	1

A	B	GT	EQ	LT
1000	0111	0	0	1
0111	1000	1	0	0
1010	1100	0	0	1
1100	1010	1	0	0
1000	1000	0	1	0

Unsigned Magnitude Multiplier Tests

A	B	Expected Product P[7:0]
0000	0000	00000000
0000	1111	00000000
0001	0001	00000001
0001	1010	00001010
0010	0010	00000100
0011	0011	00001001
0101	0011	00001111
0110	0110	00100100
1000	1000	01000000
1010	1010	01100100
1111	1111	11100001

Signed Multiplier Correction Tests

Pmag	ProductSign	Expected Final Product
00000110	0	00000110
00000110	1	11111010
00000001	1	11111111
00001000	1	11111000
00001111	1	11110001
01000000	0	01000000
01000000	1	11000000